Azure Cloud Foundations & Security Hardening

Hosting • Identity • Storage

HOSTING

IDENTITY

STORAGE

ORIGIN

Four Years Apart: AWS, Then Azure

2008
Announced as “Windows Azure” (Project Red Dog)

2010
Reaches general availability, February 1

2012
Windows Azure Websites — App Service's precursor

2014
Renamed “Microsoft Azure”

2023
Azure AD renamed Microsoft Entra ID

Sources: Microsoft Azure — Wikipedia; “Upcoming Name Change for Windows Azure” (Microsoft Azure Blog); Microsoft FY26 Q4 press release (news.microsoft.com, July 29, 2026).

The Azure Service Landscape

LANDSCAPE

A map, not a syllabus — before today narrows into three categories

Compute

App Service · VMs · AKS

TODAY

Storage

Blob · Table · Azure Files

TODAY

Databases

Azure SQL · Cosmos DB

Identity

Entra ID · RBAC

TODAY

Networking

VNet · Load Balancer

AI / Machine Learning

Azure OpenAI · ML

Security

Defender · Key Vault

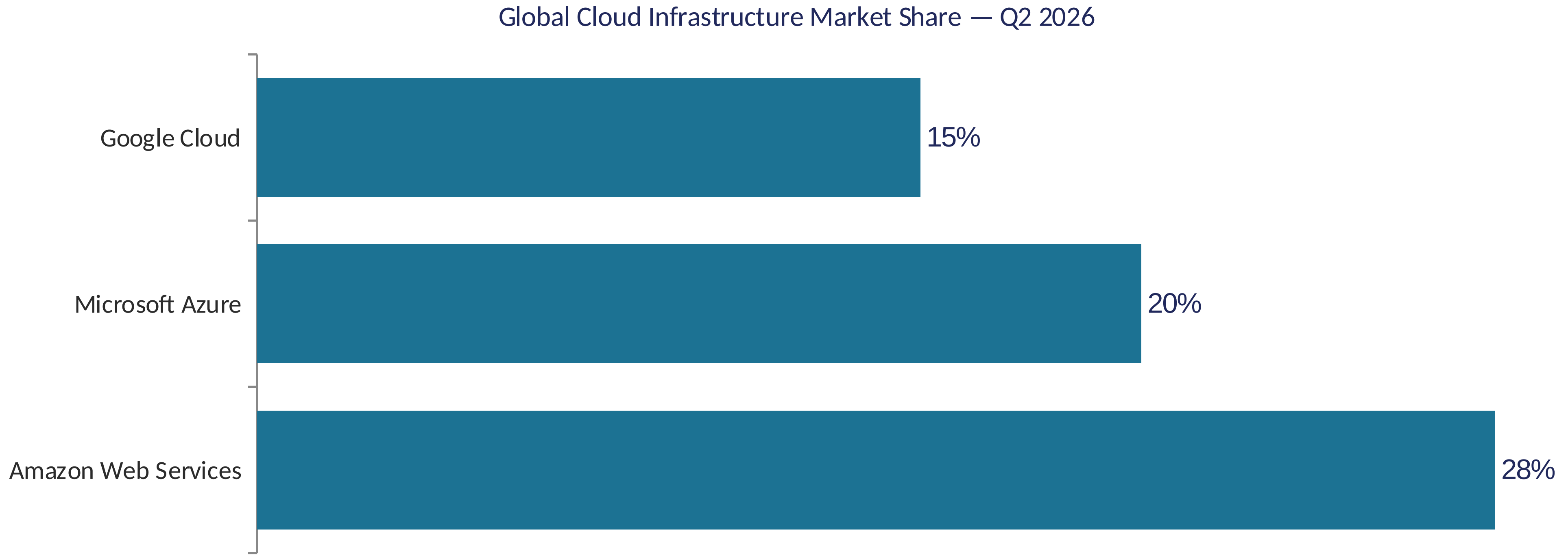
Management & Governance

Monitor · Policy

Where Azure Sits Today

COMPARATIVE

AWS leads the cloud infrastructure market, **but not by an overwhelming margin** — this course teaches both clouds because most enterprise teams work across both.



Source: Synergy Research Group, "Q2 Cloud Market Passes $143 Billion; Highest Growth Rate in Eight Years" (published July 30, 2026, srgresearch.com).

ANALOGY

Mapping Yesterday's AWS Onto Today's Azure

- Blob Storage ↔ S3 — a strong, direct analogy
- RBAC's principal/role/scope ↔ IAM's role/policy/resource — a strong, direct analogy
- App registration ↔ an EC2 instance's IAM role — a strong analogy for the problem it solves
- App Service ↔ EC2 — deliberately NOT a direct analogy: a different compute shape

S3

✓ Blob Storage

IAM role / policy / resource

✓ RBAC principal / role / scope

EC2's attached IAM role

✓ App registration

EC2 (VM you manage)

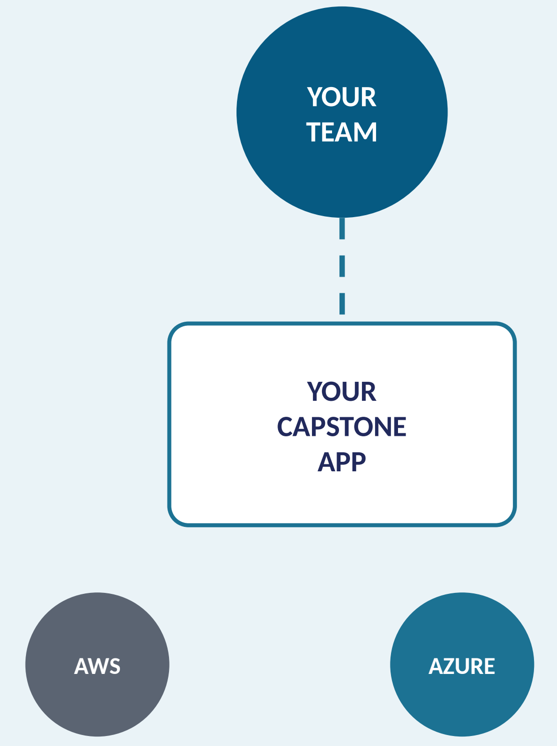
⚠️ App Service (Azure manages it)

Three clean mappings, one honest exception — flagged, not smoothed over

Your Capstone, Now on a Second Cloud

PROTAGONIST

- Everything today attaches to the same capstone application from yesterday
- The same team makes every decision here too — and reconciles both clouds later
- Every problem from here on is something your app now needs, not a new example
- Today's Azure identity gets checked against yesterday's AWS identity, not treated separately



One team, one app, now living on two clouds — carried forward, not restarted

Azure App Service & Microsoft Entra ID Essentials

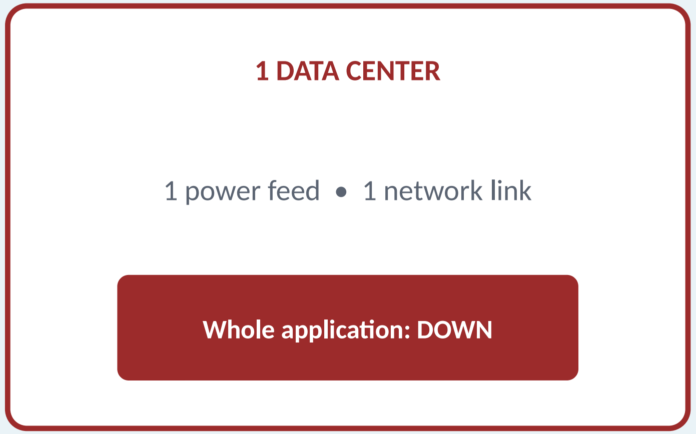
WHAT'S AHEAD

- Azure Regions & Availability Zones: where your infrastructure actually lives
- Azure App Service: managed, patch-free web app hosting
- Deployment slots: stage a release, then swap it live
- Microsoft Entra ID: identity for people, and for apps
- App registrations: give the capstone app its own identity

PROBLEM

Does Switching Clouds Erase Yesterday's Risk?

- Every hands-on step today starts by asking you to pick an Azure “Region” — before anything else
- Yesterday's single-data-center risk didn't disappear just because your app is now on a second cloud
- Azure might not use AWS's exact same terms, or the same numbers
- Picking wrong isn't free — some resources can't move Regions later, on either cloud



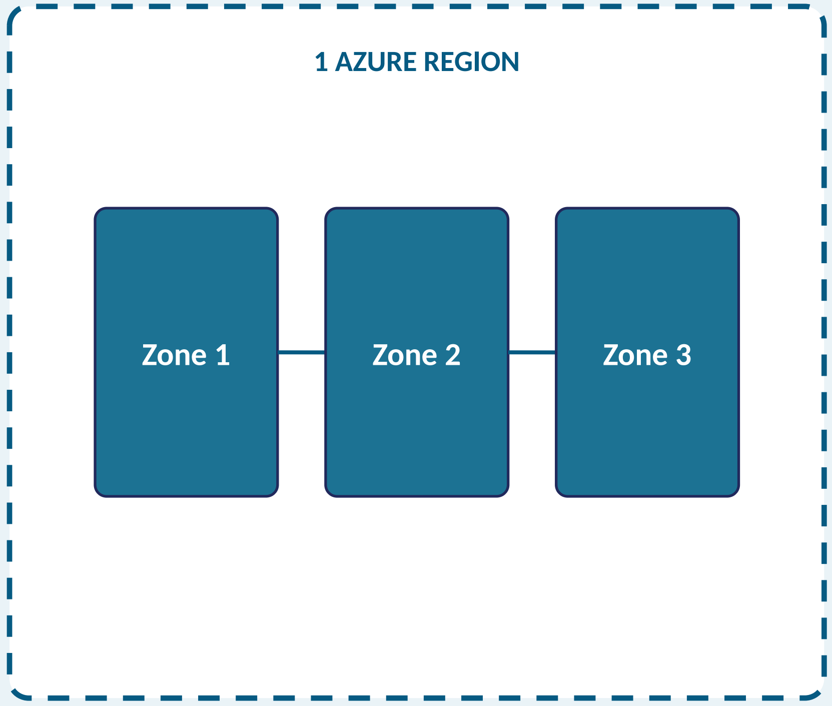
same risk, different cloud logo

The same single-data-center risk from yesterday, now on Azure's side of the capstone

SOLUTION

Azure Regions & Availability Zones

- Region: one or more datacenters linked by a high-capacity, low-latency network (80+ regions)
- Within many regions, Availability Zones add independent power, cooling, and networking
- Use two or more zones and a resource survives a single zone failure
- Region Pairs exist too, but zones, not pairs, are what actually does the resilience work
- Your Region choice today is sticky — same discipline as yesterday's AWS pick

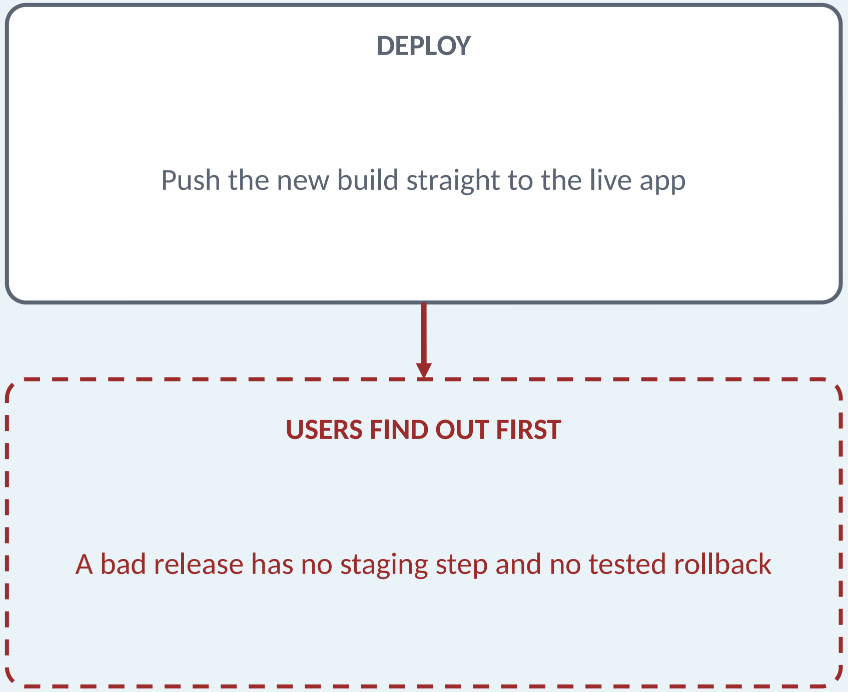


One Azure Region containing three Availability Zones — matching yesterday's AWS diagram

Hosting Without Managing the Machine

- Your app's compute right now is yesterday's EC2 instance — patched and scaled by your team
- Running a web app shouldn't require owning the machine underneath it
- Deploying straight to production risks your users finding a bad release before your team does
- No clean, tested path today to roll back if a deployment goes wrong

Today's compute is still yesterday's EC2 instance:

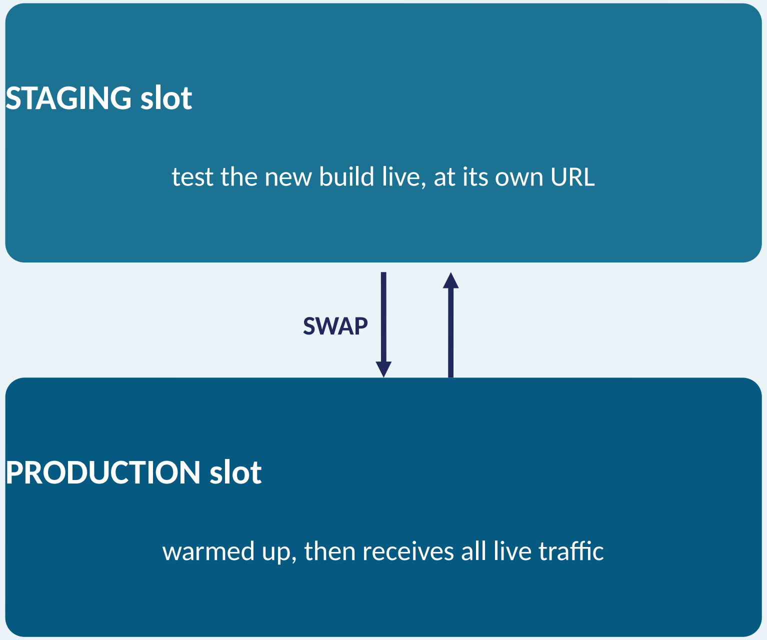


A production URL going straight from "deploy" to "users," no safety net in between

SOLUTION

Azure App Service & Deployment Slots

- Your app moves onto Azure App Service — Azure runs the VM, OS, and patching, not your team
- Supports .NET, Java, Node.js, Python, PHP, and containers, on Windows or Linux
- App Service plan sets the tier; Standard tier and up unlocks deployment slots
- A slot is a full, live copy of your app that swaps into production with no dropped requests



A staging slot swaps into production — no downtime, no dropped requests

HANDS-ON Deploy to App Service with a staging slot

PROBLEM

Identity for an Application, Not Just a Person

- Your capstone app itself now needs an identity to access Azure resources directly
- Your sign-in isn't the same thing as your application's identity
- Azure's identity service isn't IAM — different name, different shape
- Without its own identity, your app can't be granted scoped access at all

?

Whose identity is this?

A person icon and an application icon, both needing to sign in to the same resource

Microsoft Entra ID & App Registrations

SOLUTION

- Microsoft Entra ID: cloud identity and access management, formerly Azure AD
- Registering your app creates its own identity (and service principal) in the tenant
- Registration produces an Application (client) ID and a Directory (tenant) ID
- This app identity is the “who” a later role assignment grants access to

MICROSOFT ENTRA ID

cloud identity & access management

APP REGISTRATION

Application (client) ID + Directory (tenant) ID

A registration form produces two IDs: the app's own identity in the tenant

HANDS-ON Register an application in Microsoft Entra ID

Azure Storage & Cross-Cloud Security Essentials

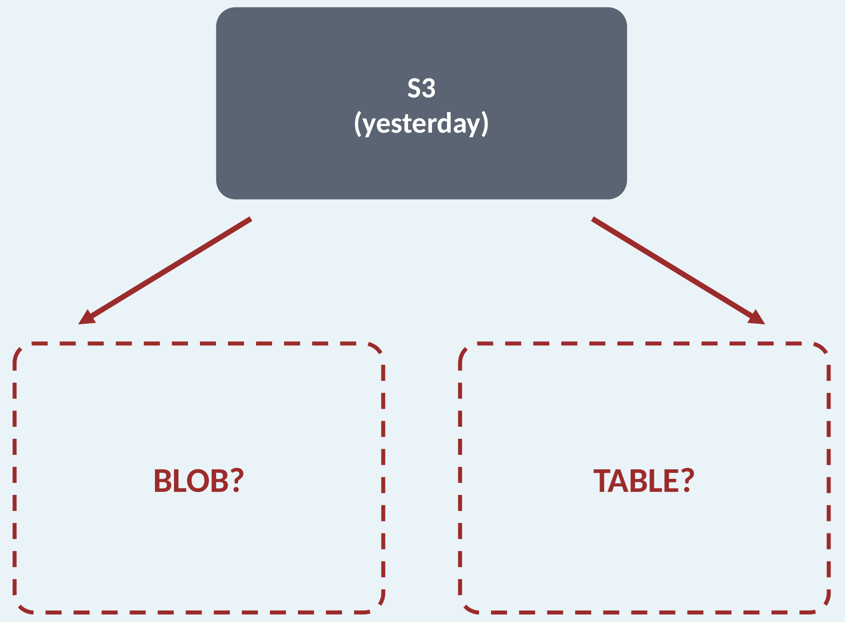
WHAT'S AHEAD

- Azure Blob storage: objects, containers, and access tiers
- Azure Table storage: schemaless structured records
- Two cloud identities now exist for your app — are both least-privilege?
- Azure RBAC compared side-by-side with AWS IAM

Same Need, Which Azure Service?

PROBLEM

- Your capstone needs somewhere to put data on its second cloud too
- Yesterday's answer was one service, S3, for one shape of data
- Azure splits that same need across two different, purpose-built services
- Picking the wrong one means fighting the service's shape later



Yesterday's one storage service branches into two purpose-built choices today

Azure Blob Storage & Table Storage

SOLUTION

- Blob storage: unstructured objects — files, images, logs, backups, same shape as yesterday's S3
- Storage account → container → blob; block, append, and page blob types
- Access tiers: hot, cool, cold, archive — cost trades against retrieval speed
- Table storage: schemaless NoSQL records, keyed by PartitionKey and RowKey

BLOB STORAGE

account → container → blob • hot/cool/cold/archive

TABLE STORAGE

table → entity • PartitionKey + RowKey

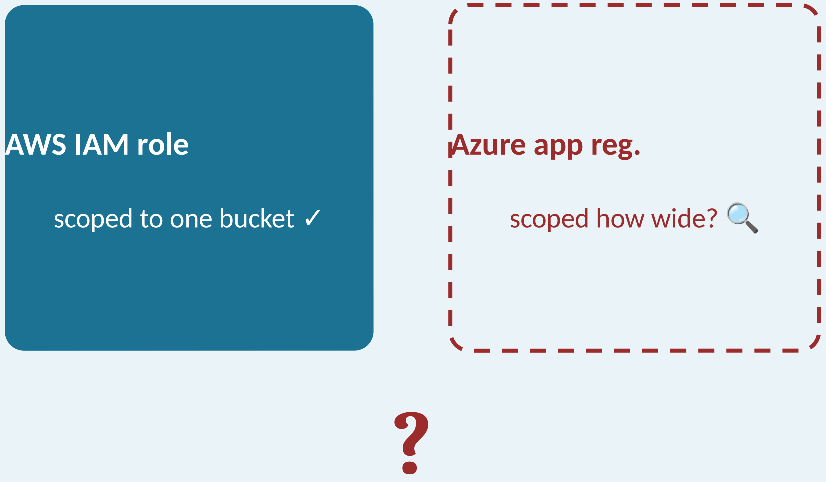
Same storage account, two purpose-built shapes: objects and structured records

HANDS-ON Create Blob and Table storage for the capstone

Two Clouds, Two Identities, One Blind Spot

PROBLEM

- Your capstone app now has an identity on AWS (yesterday) and on Azure (today)
- Getting one identity's permissions right doesn't guarantee the other is too
- A broad grant on one cloud can go unnoticed while the other stays tight
- Without comparing both, an audit or incident is the first time anyone checks



One identity confirmed least-privilege; the other, unchecked, until now

Azure RBAC, Compared to AWS IAM

SOLUTION

- Role assignment = principal (who) + role (what) + scope (where)
- Principal: your app registration · Role: Storage Blob Data Contributor · Scope: one storage account
- Best practice: smallest scope needed — same discipline as naming one bucket's ARN
- IAM role ≈ app registration; IAM policy ≈ role definition; attachment ≈ scope

AWS IAM	Azure RBAC
IAM role	App registration
IAM policy	Role definition
Resource ARN	Scope

Principal: your app registration
Role: `Storage Blob Data Contributor`
Scope: this one storage account

Same least-privilege discipline — the concrete values, next to each cloud's own vocabulary

HANDS-ON Write a least-privilege RBAC set across both clouds